

DIGITAL TO ANALOG CONVERSION

Reference book: Advanced Electronic Communications Systems by Wayne Tomasi (Sixth Edition)

INTRODUCTION

❖ Digital to Analog Conversion:

1. Amplitude Shift Keying(ASK)
2. Frequency Shift Keying(FSK)
3. Phase Shift Keying(PSK)
4. Quadrature Amplitude Modulation(QAM)

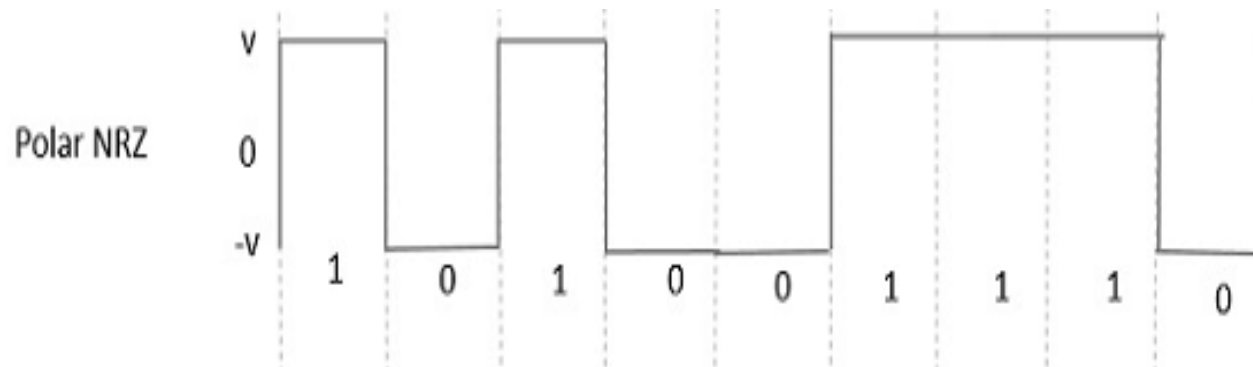


Fig: Polar Return to Zero(NRZ) Signal

AMPLITUDE SHIFT KEYING

- Amplitude-shift keying is sometimes called digital amplitude modulation (DAM). Mathematically, amplitude-shift keying is

$$v_{(ask)}(t) = [1 + v_m(t)] \left[\frac{A}{2} \cos(\omega_c t) \right]$$

Where,

$v_{ask}(t)$ = amplitude-shift keying wave

$v_m(t)$ = digital information (modulating) signal (volts)

$A/2$ = unmodulated carrier amplitude (volts)

ω_c = analog carrier radian frequency (radians per second, $2\pi f_c t$)

- The modulating signal ($v_m(t)$) is a normalized binary waveform, where $+1\text{ V} = \text{logic 1}$ and $-1\text{ V} = \text{logic 0}$

- Therefore, for a logic 1 input, $v_m(t) = +1\text{ V}$,
$$v_{(ask)}(t) = [1 + 1] \left[\frac{A}{2} \cos(\omega_c t) \right]$$
$$= A \cos(\omega_c t)$$

- and for a logic 0 input, $v_m(t) = -1\text{ V}$,
$$v_{(ask)}(t) = [1 - 1] \left[\frac{A}{2} \cos(\omega_c t) \right]$$
$$= 0$$

AMPLITUDE SHIFT KEYING CONT'D

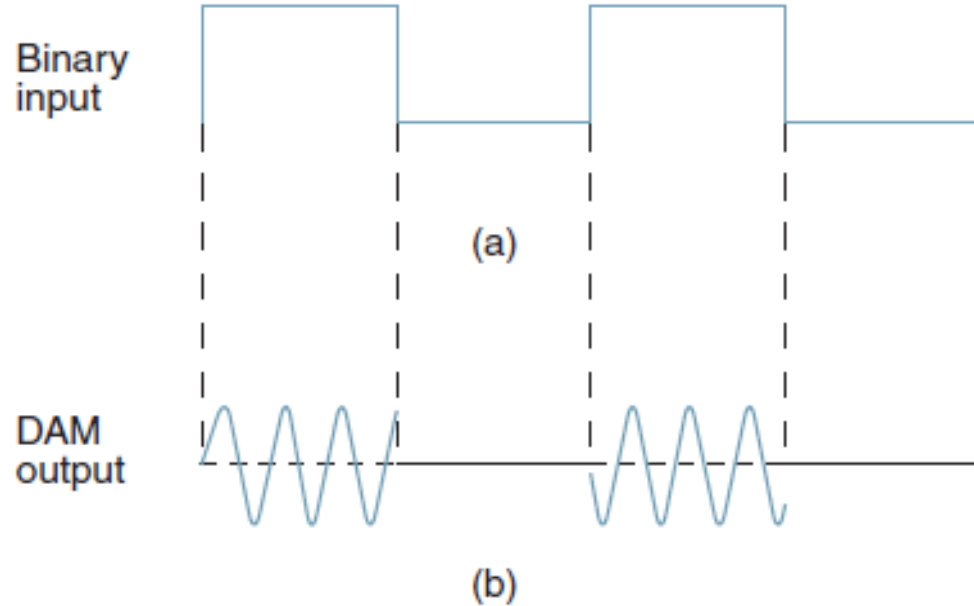


Fig: Digital amplitude modulation: (a) input binary; (b) output DAM waveform

- Hence, the carrier is either “on” or “off,” which is why amplitude-shift keying is sometimes referred to as on-off keying (OOK).

FREQUENCY SHIFT KEYING

- FSK is sometimes called binary FSK (BFSK). The general expression for FSK is

$$v_{fsk}(t) = V_c \cos\{2\pi[f_c + v_m(t) \Delta f]t\}$$

Where,

$v_{fsk}(t)$ = binary FSK waveform

V_c = peak analog carrier amplitude (volts)

f_c = analog carrier center frequency (hertz)

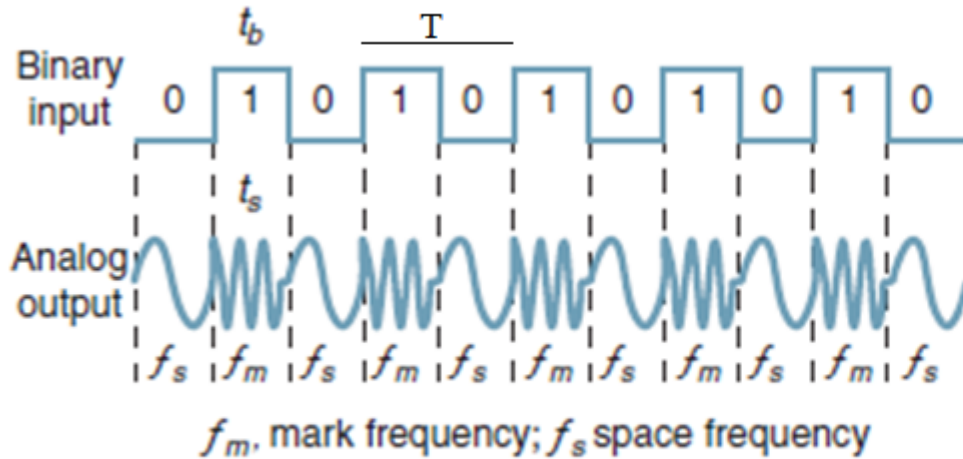
Δf = peak change (shift) in the analog carrier frequency (hertz)

$v_m(t)$ = binary input (modulating) signal (volts)

- Therefore, for a logic 1 input, $v_m(t) = +1$ V, $v_{fsk}(t) = V_c \cos[2\pi(f_c + \Delta f)t]$

- and for a logic 0 input, $v_m(t) = -1$ V, $v_{fsk}(t) = V_c \cos[2\pi(f_c - \Delta f)t]$

FREQUENCY SHIFT KEYING CONT'D



binary input	frequency output
0	space (f_s)
1	mark (f_m)

(a)

(b)

Fig: FSK in the time domain: (a) waveform; (b) truth table

Mark frequency, $f_m = f_c + \Delta f$

Space frequency, $f_s = f_c - \Delta f$

Frequency deviation (hertz), $\Delta f = \frac{|f_m - f_s|}{2}$

Highest fundamental frequency of the binary input signal, $f_a = \frac{1}{T}$

T sec ----- 2 bit

1 sec ----- $\frac{2}{T}$ bit/sec

Input bit rate (bps), $f_b = \frac{2}{T} = 2f_a$

FREQUENCY SHIFT KEYING CONT'D

- Modulation index, $h = \frac{\Delta f}{f_a} = \frac{\frac{(f_m - f_s)}{2}}{f_a} = \frac{(f_m - f_s)}{2f_a} = \frac{(f_m - f_s)}{f_b}$

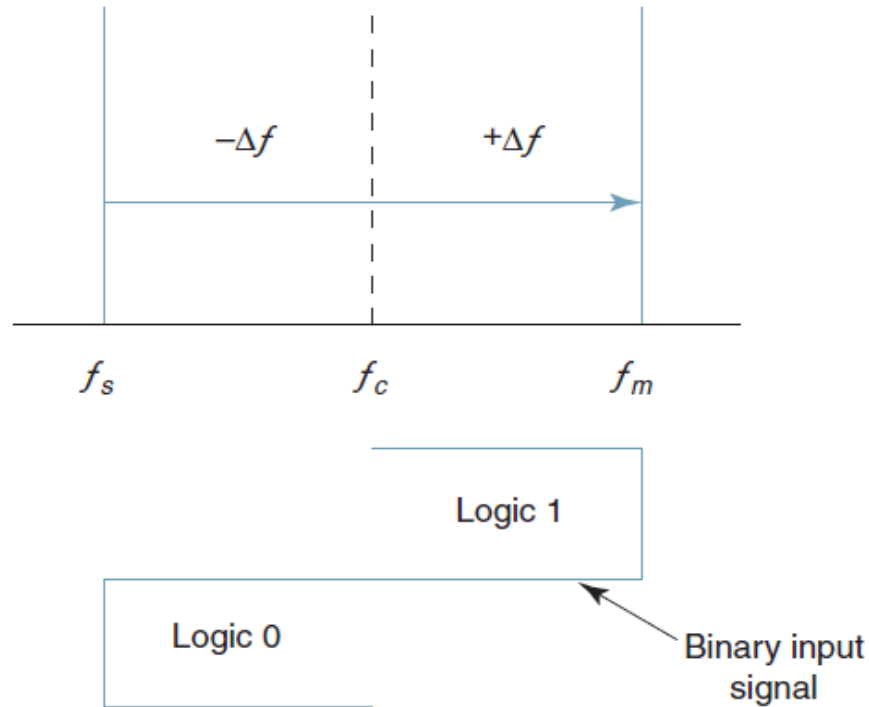


Fig: FSK in the frequency domain

FREQUENCY SHIFT KEYING CONT'D

❖ FSK Transmitter:

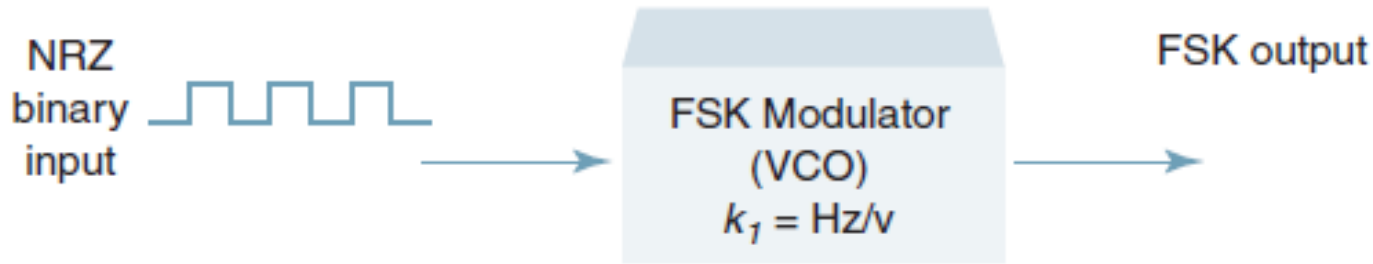


Fig: FSK modulator

- A logic 1 input shifts the voltage-controlled oscillator (VCO) output to the mark frequency, and a logic 0 input shifts the VCO output to the space frequency.
- Consequently, as the binary input signal changes back and forth between logic 1 and logic 0 conditions, the VCO output shifts or deviates back and forth between the mark and space frequencies.

FREQUENCY SHIFT KEYING CONT'D

❖ FSK Receiver:

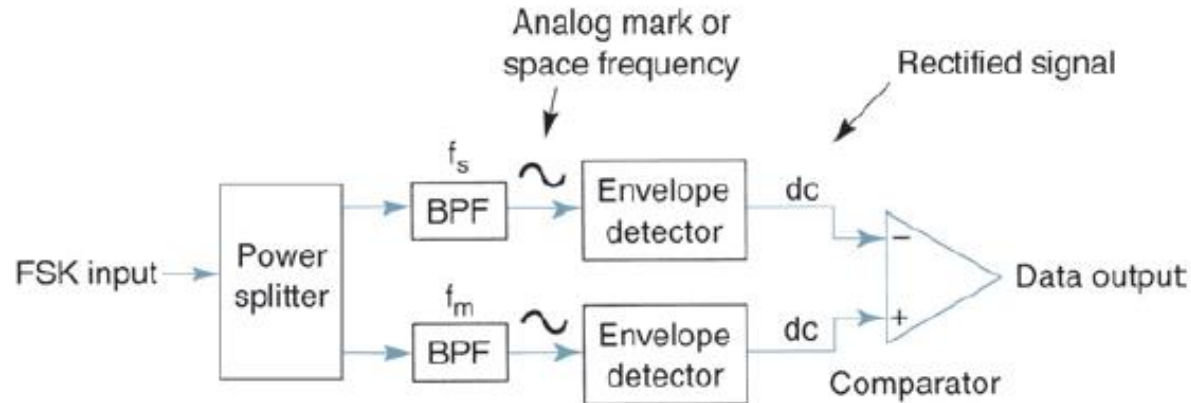


Fig: Noncoherent FSK demodulator

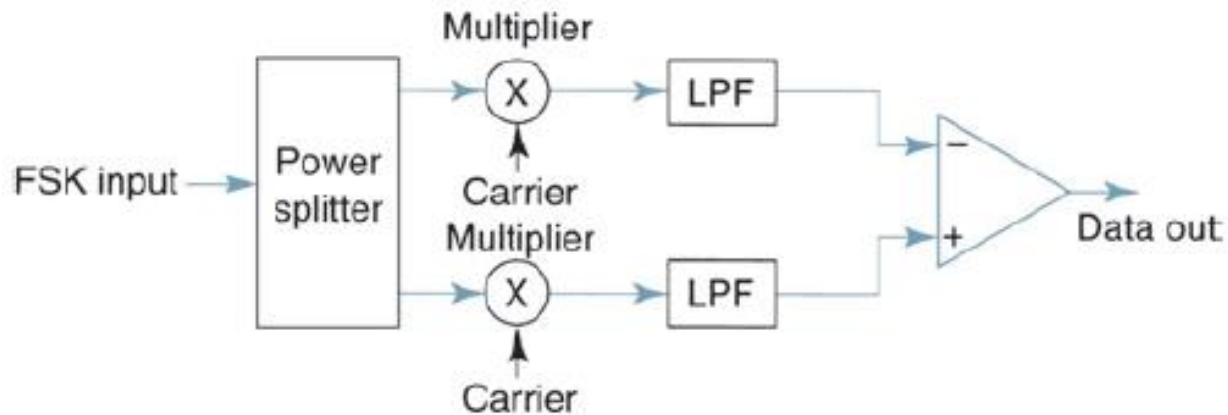


Fig: Coherent FSK demodulator

FREQUENCY SHIFT KEYING CONT'D

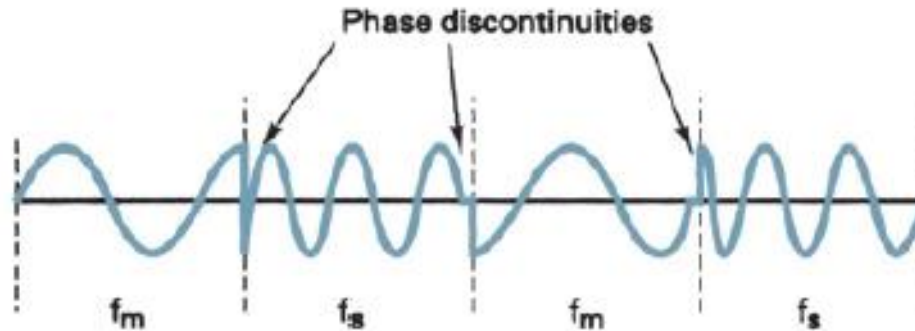


Fig: Noncontinuous FSK waveform

❖ Limitation:

- It can be seen that when the input changes from a logic 1 to a logic 0 and vice versa, there is an abrupt phase discontinuity in the analog signal.
- When this occurs, the demodulator has trouble following the frequency shift; consequently, an error may occur.

CONTINUOUS-PHASE FREQUENCY-SHIFT KEYING

- With CP-FSK, the mark and space frequencies are selected such that they are separated from the center frequency by an exact multiple of one-half the bit rate (f_m and $f_s = n[f_b/2]$), where n = any integer).

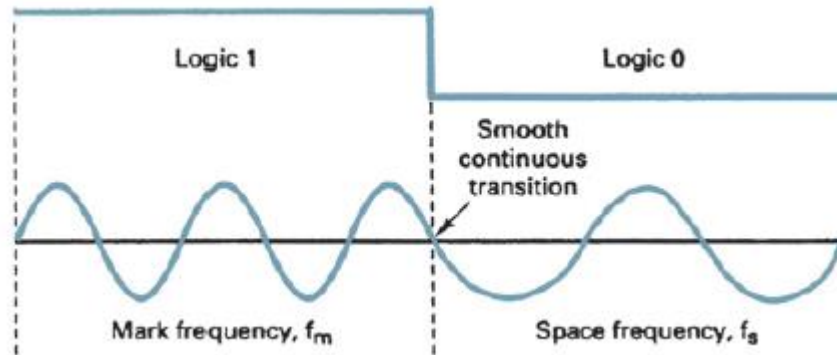


Fig: CP-FSK waveform

- When the output frequency changes, it is a smooth, continuous transition. Consequently, there are no phase discontinuities.
- CP-FSK has a better bit-error performance than conventional binary FSK for a given signal-to-noise ratio.
- The disadvantage of CP-FSK is that it requires synchronization circuits and is, therefore, more expensive to implement.

PHASE SHIFT KEYING

- The simplest form of PSK is binary phase-shift keying (BPSK).

- $$\text{PSK} = \begin{cases} \text{Logic 1} \rightarrow \sin(2\pi f_c t) \\ \text{Logic 0} \rightarrow \sin(2\pi f_c t + \pi) \\ \quad \text{or } -\sin(2\pi f_c t) \end{cases}$$

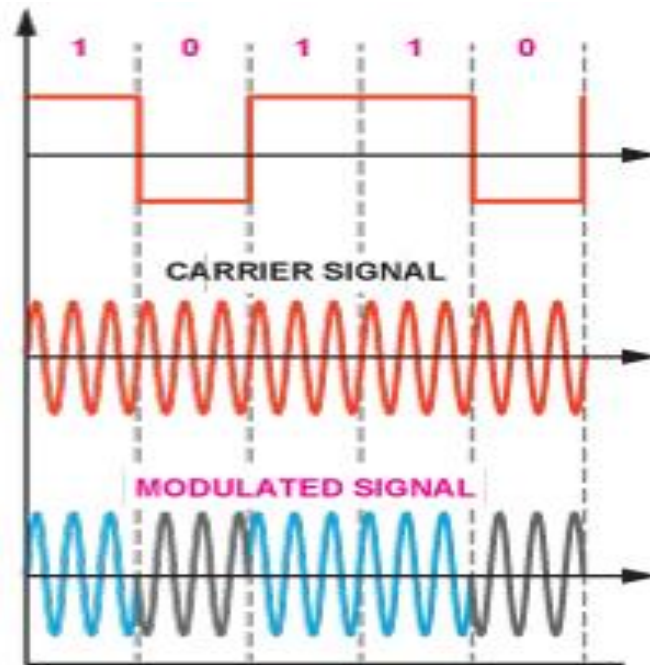


Fig: Waveform of PSK

PHASE SHIFT KEYING CONT'D

- BPSK transmitter:

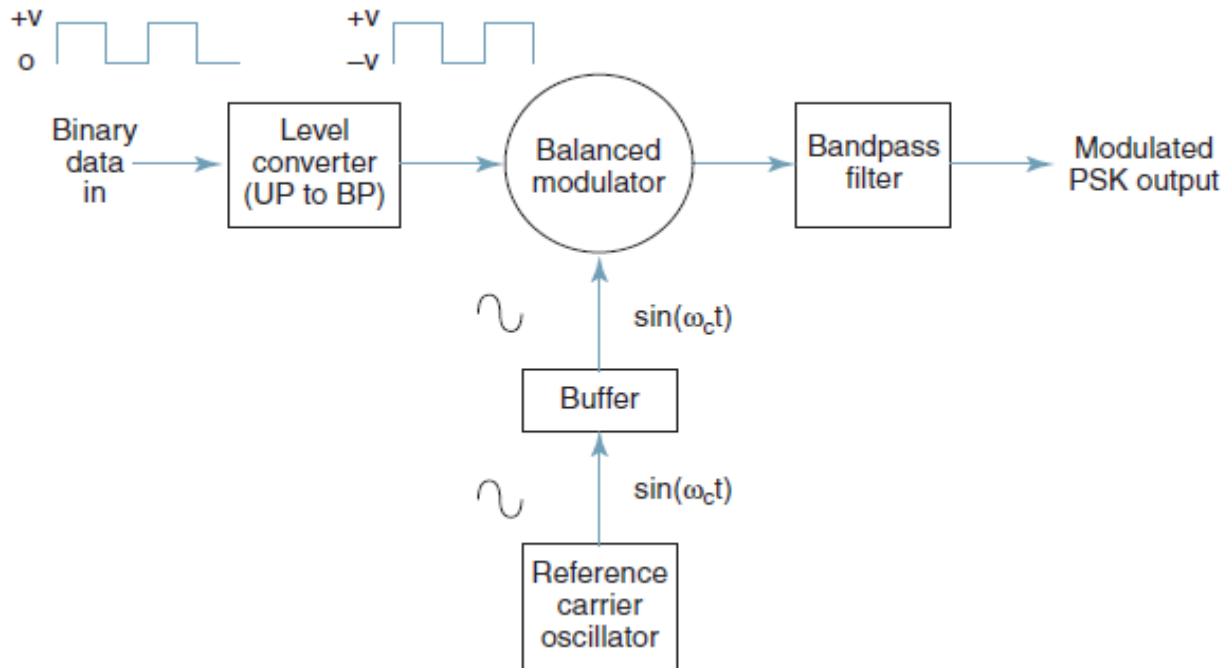


Fig: BPSK transmitter

PHASE SHIFT KEYING CONT'D

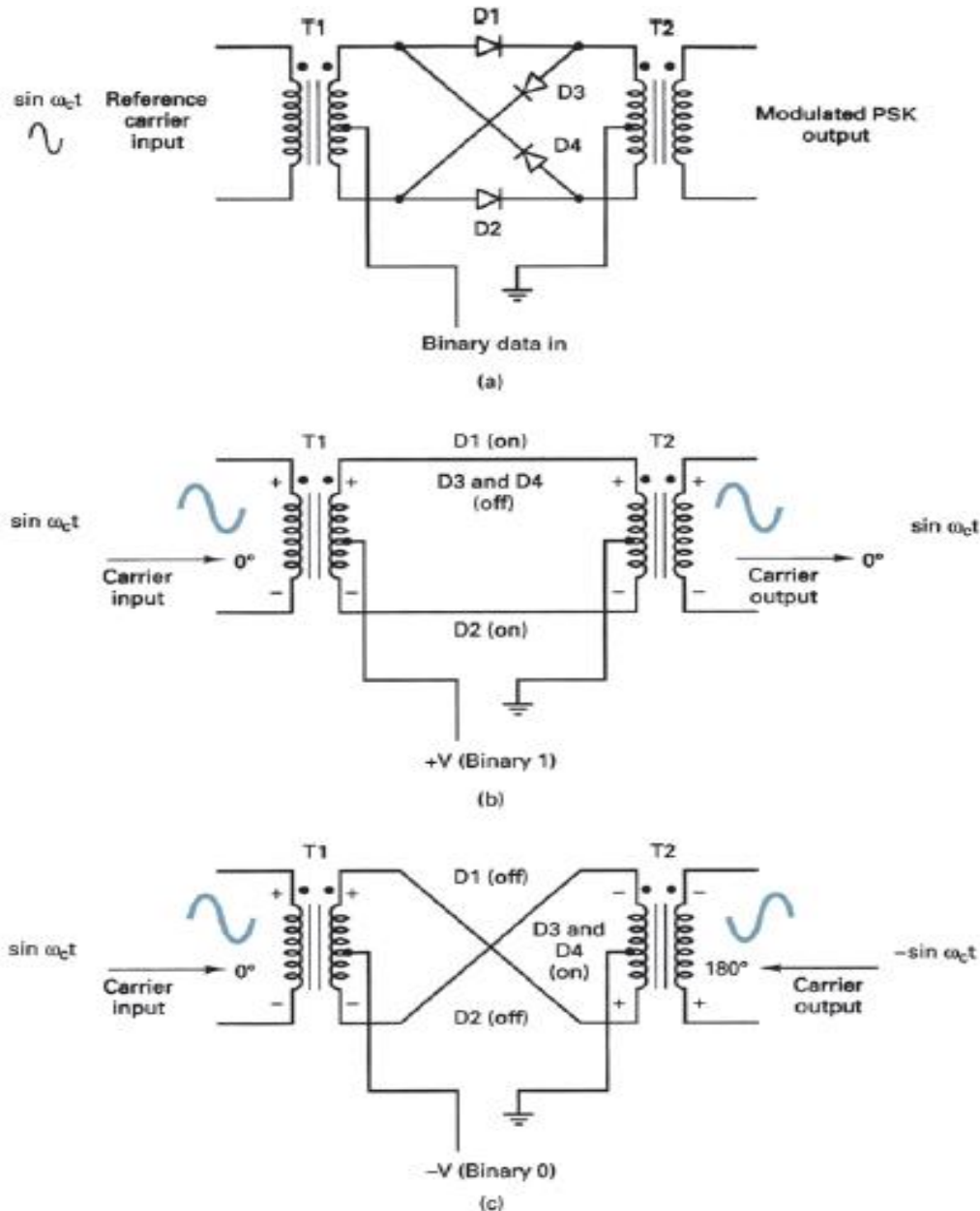


Fig: (a) Balanced ring modulator; (b) logic 1 input; (c) logic 0 input

PHASE SHIFT KEYING CONT'D

❖ **Bandwidth considerations of BPSK:**

Mathematically, the output of the balanced modulators is $\theta = (\sin \omega_a t)(\sin \omega_c t)$
 $= (\sin 2\pi f_a t)(\sin 2\pi f_c t)$

$$= \frac{1}{2} [\cos 2\pi(f_c - f_a)t - \cos 2\pi(f_c + f_a)t]$$

Minimum bandwidth is $(f_c + f_a) - (f_c - f_a) = 2f_a$

$$= 2 \frac{f_b}{2} \quad [\text{N.B: } f_a = \frac{\text{Bit rate}}{2}]$$
$$= f_b$$

PHASE SHIFT KEYING CONT'D

○ BPSK receiver:

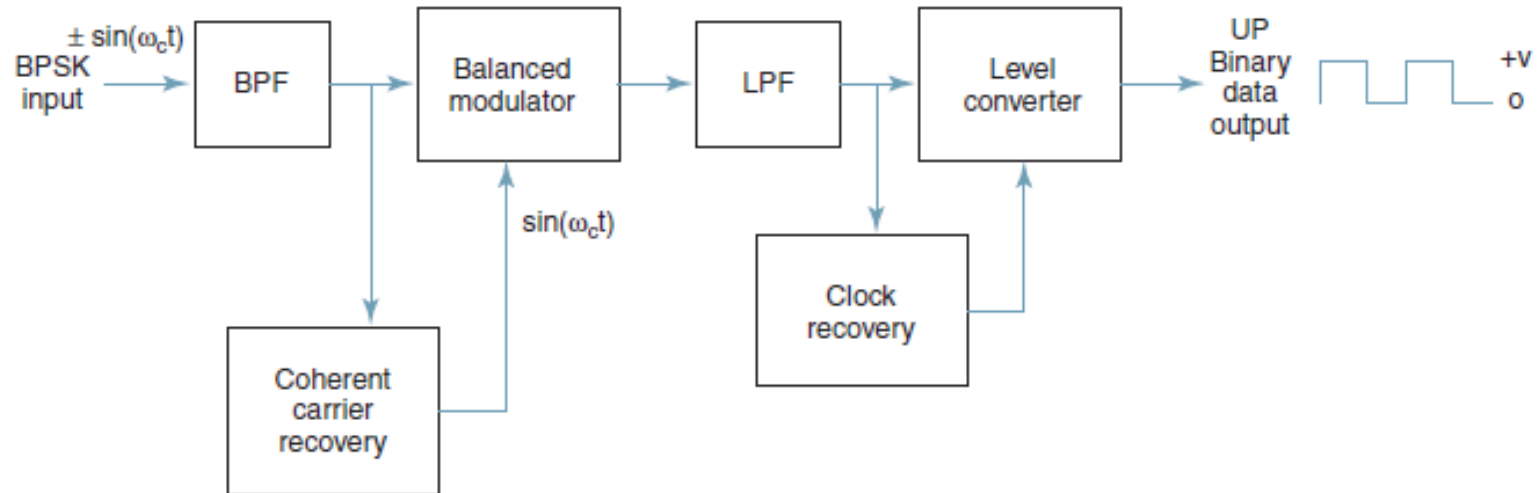


Fig: BPSK receiver

○ For input signal of logic 1,

$$\text{output} = (\sin \omega_c t)(\sin \omega_c t) = \sin^2 \omega_c t$$

or

$$\sin^2 \omega_c t = \frac{1}{2}(1 - \cos 2\omega_c t) = \frac{1}{2} - \frac{1}{2} \overset{\text{(filtered out)}}{\cos 2\omega_c t}$$

leaving

$$\text{output} = +\frac{1}{2}V = \text{logic 1}$$

PHASE SHIFT KEYING CONT'D

○ BPSK receiver:

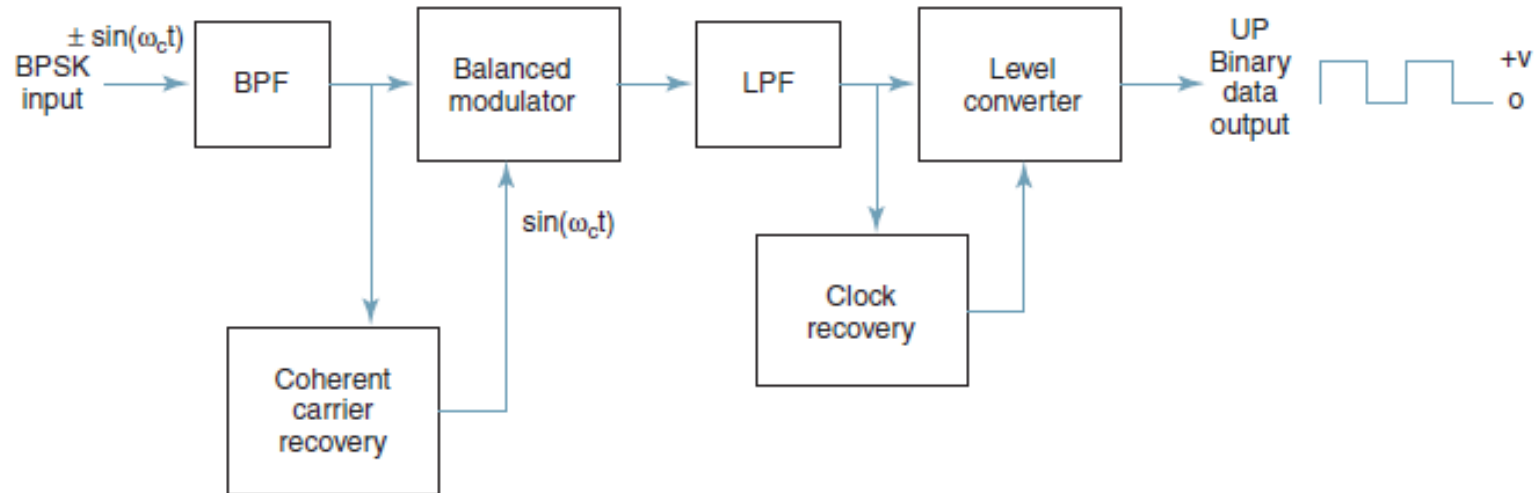


Fig: BPSK receiver

○ For input signal of logic 0,

$$\text{output} = (-\sin \omega_c t)(\sin \omega_c t) = -\sin^2 \omega_c t$$

$$\text{or} \quad -\sin^2 \omega_c t = -\frac{1}{2}(1 - \cos 2\omega_c t) = -\frac{1}{2} + \frac{1}{2} \overset{\text{(filtered out)}}{\cos 2\omega_c t}$$

$$\text{leaving} \quad \text{output} = -\frac{1}{2}V = \text{logic 0}$$

QUATERNARY PHASE-SHIFT KEYING

- Quaternary phase shift keying (QPSK), or quadrature PSK as it is sometimes called, is another form of angle-modulated, constant-amplitude digital modulation.
- QPSK is an M-ary encoding scheme where $N = 2$ and $M = 4$ (hence, the name “quaternary” meaning “4”).
- With QPSK, four output phases are possible for a single carrier frequency.
- Because there are four output phases, there must be four different input conditions.
- With two bits, there are four possible conditions: 00, 01, 10, and 11. Therefore, with QPSK, The binary input data are combined into groups of two bits, called dibits.

QUATERNARY PHASE-SHIFT KEYING CONT'D

❖ QPSK transmitter:

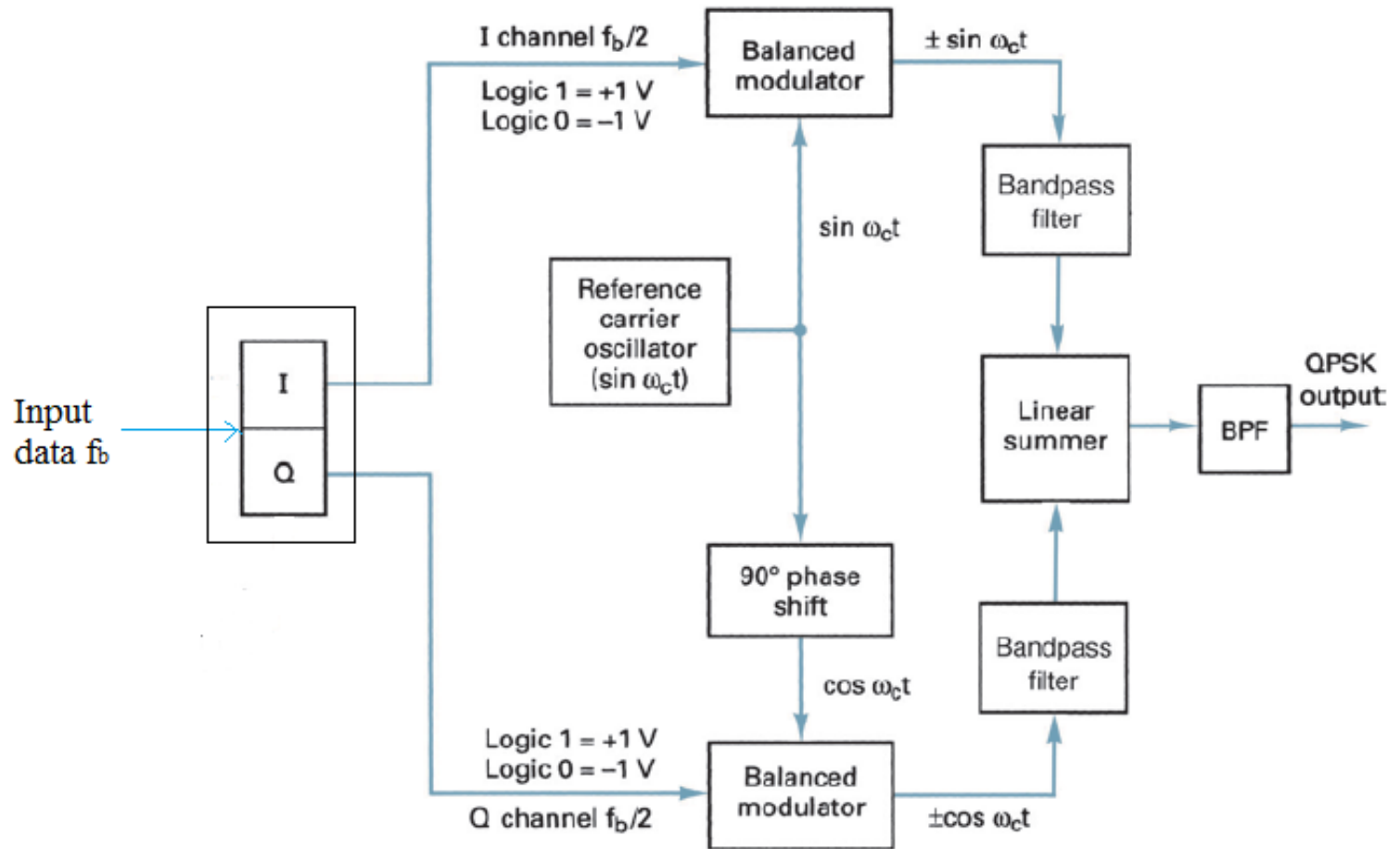


Fig: QPSK modulator

- The I bit modulates a carrier that is in phase with the reference oscillator (hence the name “I” for “in phase” channel), and the Q bit modulates a carrier that is 90° out of phase or in quadrature with the reference carrier (hence the name “Q” for “quadrature” channel).

QUATERNARY PHASE-SHIFT KEYING CONT'D

- ❖ **Example:** For a binary data input of $Q = 0$ and $I = 0$, the two inputs to the I balanced modulator are -1 and $\sin \omega_c t$, and the two inputs to the Q balanced modulator are -1 and $\cos \omega_c t$. Consequently, the outputs are

$$\text{I balanced modulator} = (-1)(\sin \omega_c t) = -\sin \omega_c t$$

$$\text{Q balanced modulator} = (-1)(\cos \omega_c t) = -\cos \omega_c t$$

and the output of the linear summer is

$$-\sin \omega_c t - \cos \omega_c t$$

$$= \sqrt{2} \left[-\frac{1}{\sqrt{2}} \sin \omega_c t - \frac{1}{\sqrt{2}} \cos \omega_c t \right] \quad [\text{N.B.: } \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}]$$

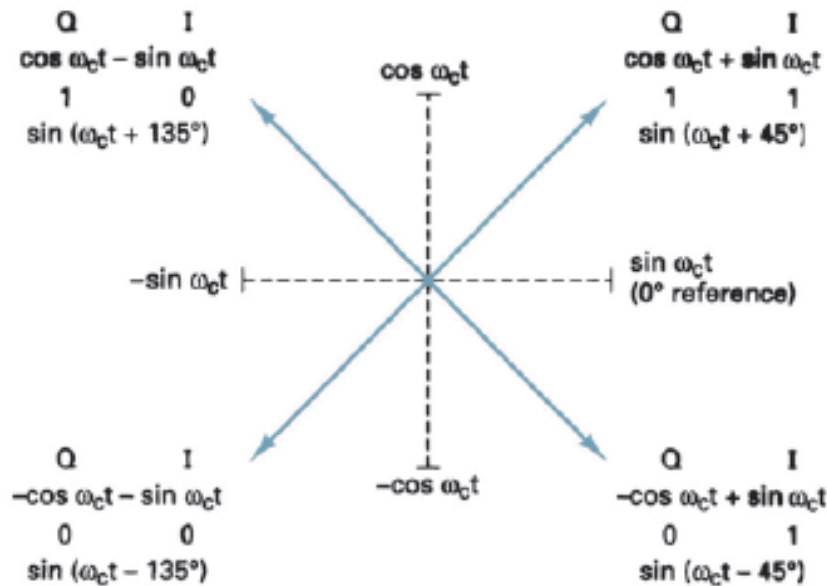
$$= \sqrt{2} [\sin \omega_c t \cos 135^\circ - \sin 135^\circ \cos \omega_c t]$$

$$= 1.41 \sin(\omega_c t - 135^\circ)$$

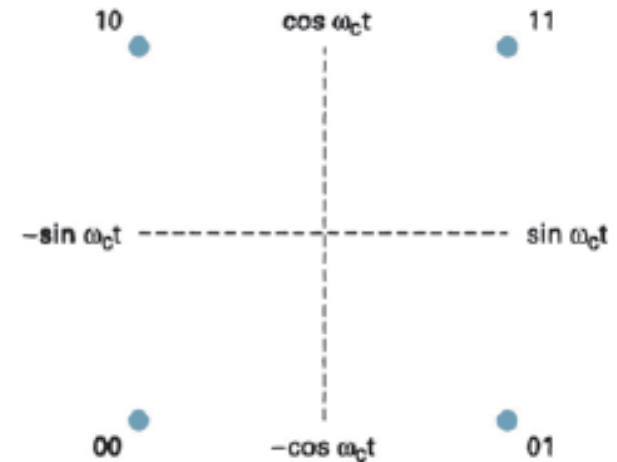
QUATERNARY PHASE-SHIFT KEYING CONT'D

Binary input		QPSK output phase
Q	I	
0	0	-135°
0	1	-45°
1	0	$+135^\circ$
1	1	$+45^\circ$

(a)



(b)



(c)

Fig: QPSK modulator: (a) truth table; (b) phasor diagram; (c) constellation diagram

QUATERNARY PHASE-SHIFT KEYING CONT'D

Bandwidth considerations of QPSK:

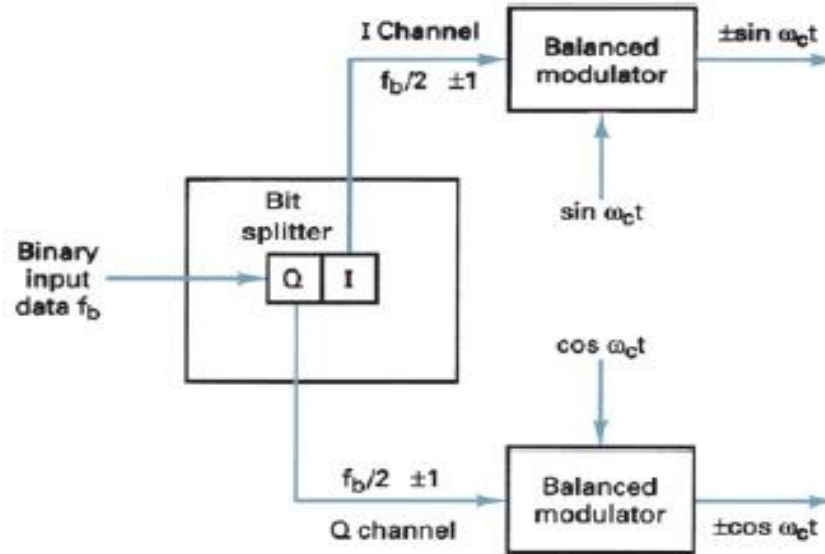


Fig: Bandwidth considerations of a QPSK modulator

Mathematically, the output of the balanced modulators is $\theta = (\sin \omega_a t)(\sin \omega_c t)$

$$= (\sin 2\pi f_a t)(\sin 2\pi f_c t)$$

$$= \frac{1}{2} [\cos 2\pi(f_c - f_a)t - \cos 2\pi(f_c + f_a)t]$$

Minimum bandwidth is $(f_c + f_a) - (f_c - f_a) = 2f_a$

$$= 2 \frac{f_b/2}{2} \quad [\text{N.B: } f_a = \frac{\text{Bit rate}}{2}]$$

$$= \frac{f_b}{2}$$

QUATERNARY PHASE-SHIFT KEYING CONT'D

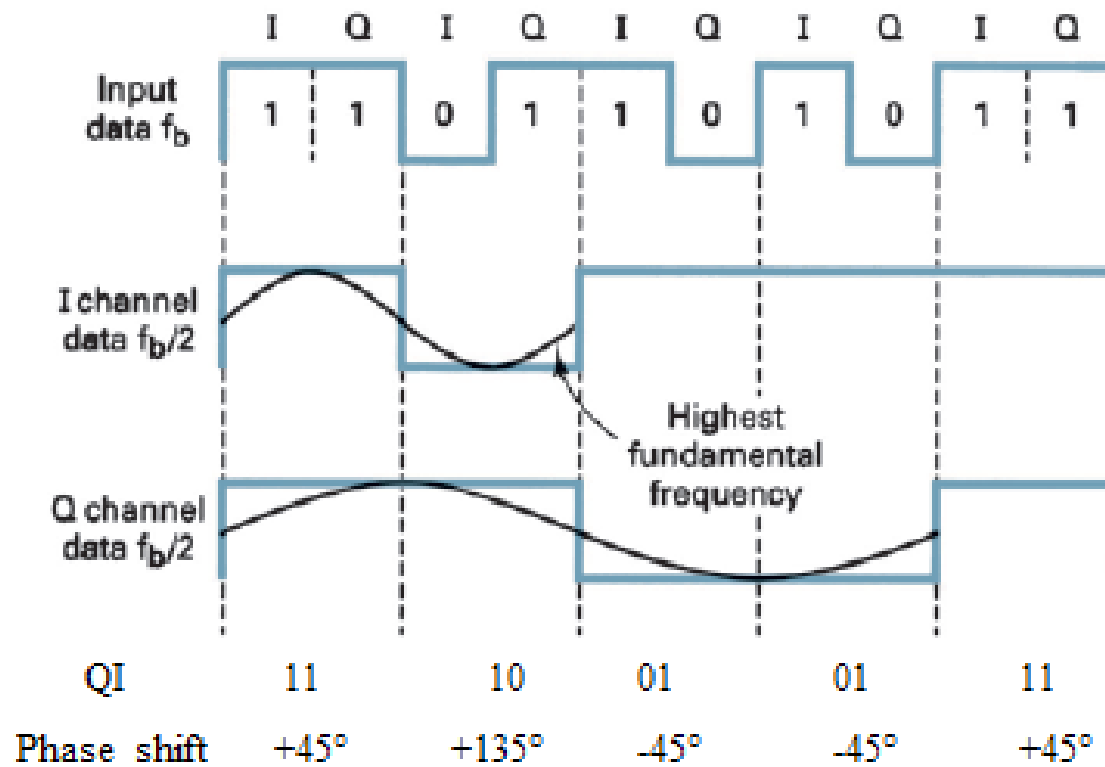


Fig: Timing diagram of QPSK

QUATERNARY PHASE-SHIFT KEYING CONT'D

❖ QPSK receiver:

- The receive QPSK signal $(-\sin \omega_c t + \sin \omega_c t)$ is one of the inputs to the I product detector. The other input is the recovered carrier $(\sin \omega_c t)$. The output of the I product detector is

$$\begin{aligned} I &= \underbrace{(-\sin \omega_c t + \cos \omega_c t)}_{\text{QPSK input signal}} \underbrace{(\sin \omega_c t)}_{\text{carrier}} \\ &= (-\sin \omega_c t)(\sin \omega_c t) + (\cos \omega_c t)(\sin \omega_c t) \\ &= -\sin^2 \omega_c t + (\cos \omega_c t)(\sin \omega_c t) \\ &= -\frac{1}{2}(1 - \cos 2\omega_c t) + \frac{1}{2}\sin(\omega_c + \omega_c)t + \frac{1}{2}\sin(\omega_c - \omega_c)t \\ I &= -\frac{1}{2} + \frac{1}{2}\cos 2\omega_c t + \frac{1}{2}\sin 2\omega_c t + \frac{1}{2}\sin 0 \\ &\quad \begin{array}{l} \nearrow \text{(filtered out)} \\ \nearrow \text{(equals 0)} \end{array} \\ &= -\frac{1}{2} \text{V (logic 0)} \end{aligned}$$

QUATERNARY PHASE-SHIFT KEYING CONT'D

Again, the receive QPSK signal $(-\sin \omega_c t + \cos \omega_c t)$ is one of the inputs to the Q product detector. The other input is the recovered carrier shifted 90° in phase $(\cos \omega_c t)$. The output of the Q product detector is

$$\begin{aligned} Q &= \underbrace{(-\sin \omega_c t + \cos \omega_c t)}_{\text{QPSK input signal}} \underbrace{(\cos \omega_c t)}_{\text{carrier}} \\ &= \cos^2 \omega_c t - (\sin \omega_c t)(\cos \omega_c t) \\ &= \frac{1}{2}(1 + \cos 2\omega_c t) - \frac{1}{2}\sin(\omega_c + \omega_c)t - \frac{1}{2}\sin(\omega_c - \omega_c)t \\ Q &= \frac{1}{2} + \frac{1}{2}\cos 2\omega_c t \overset{\text{(filtered out)}}{\nearrow} - \frac{1}{2}\sin 2\omega_c t \overset{\text{(equals 0)}}{\nearrow} - \frac{1}{2}\sin 0 \\ &= \frac{1}{2}V(\text{logic 1}) \end{aligned}$$

OFFSET QUATERNARY PHASE-SHIFT KEYING

Problems of QPSK:

In conventional QPSK, a change in the input dibit from 00 to 11 or 01 to 10 causes a corresponding 180° shift in the output phase. Such large phase transition over a small symbol interval causes momentary but large amplitude change in the signal. This leads to relatively higher sidelobe peaks in the spectrum and it is avoidable to a considerable extent by adopting a simple trick.

- Offset QPSK (OQPSK) is a modified form of QPSK where the bit waveforms on the I and Q channels are offset or shifted in phase from each other by one-half of a bit time.
- Because changes in the I channel occur at the midpoints of the Q channel bits and vice versa, there is never more than a single bit change in the dibit code and, therefore, there is never more than a 90° shift in the output phase.
- The resultant effect is that the sidelobe levels decrease to a good extent. This simple and practical variation of QPSK is known as Offset QPSK.

OFFSET QUATERNARY PHASE-SHIFT KEYING

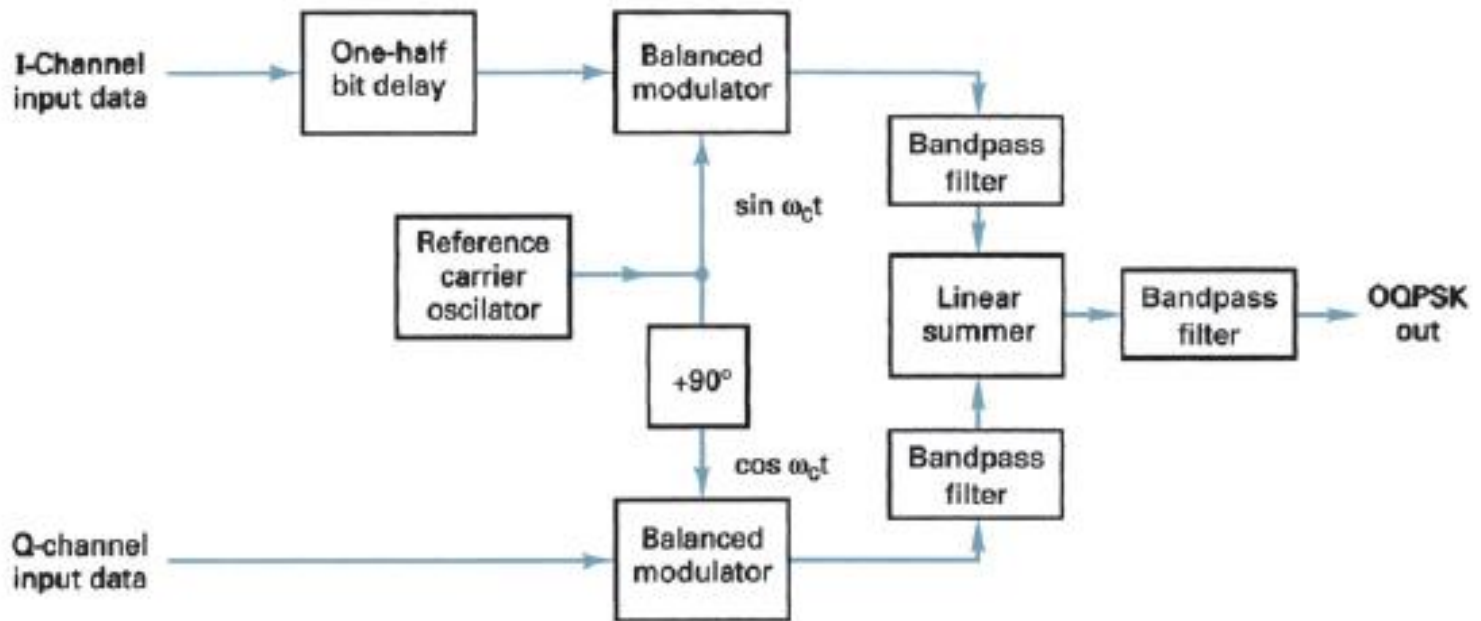


Fig: Block diagram of Offset keyed (OQPSK)

OFFSET QUATERNARY PHASE-SHIFT KEYING

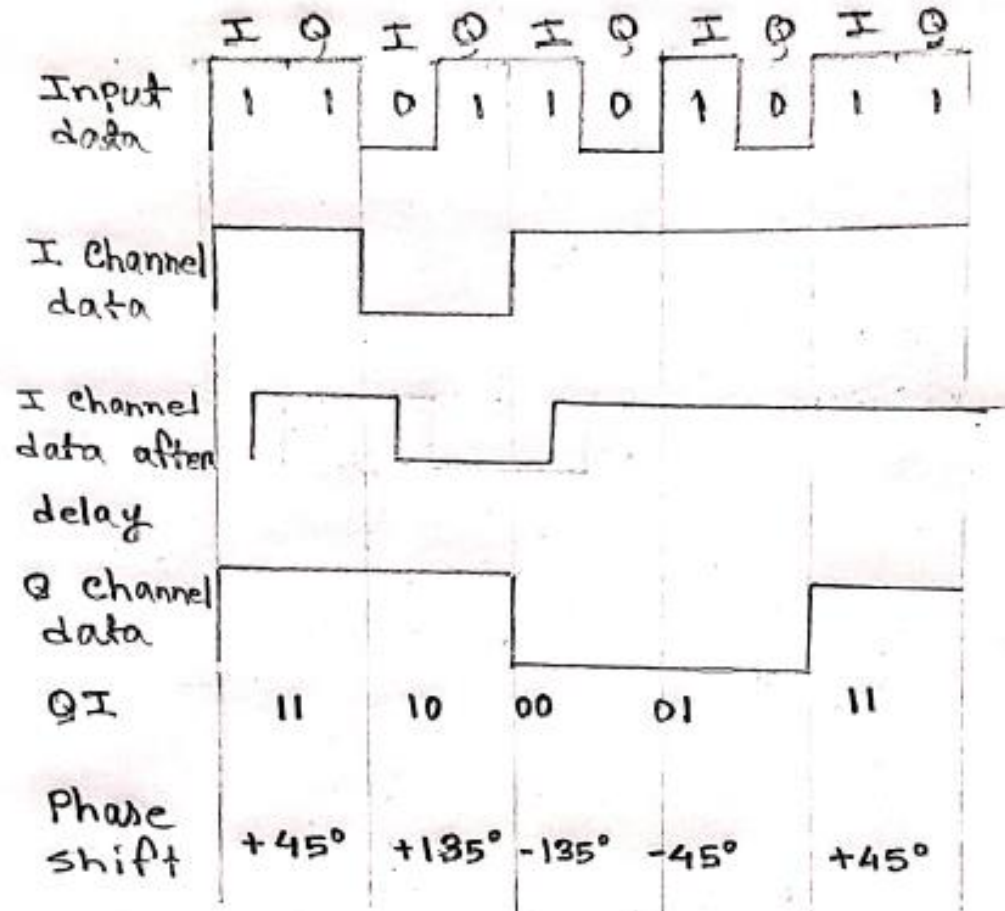


Fig: Timing diagram of OQPSK

8-PSK

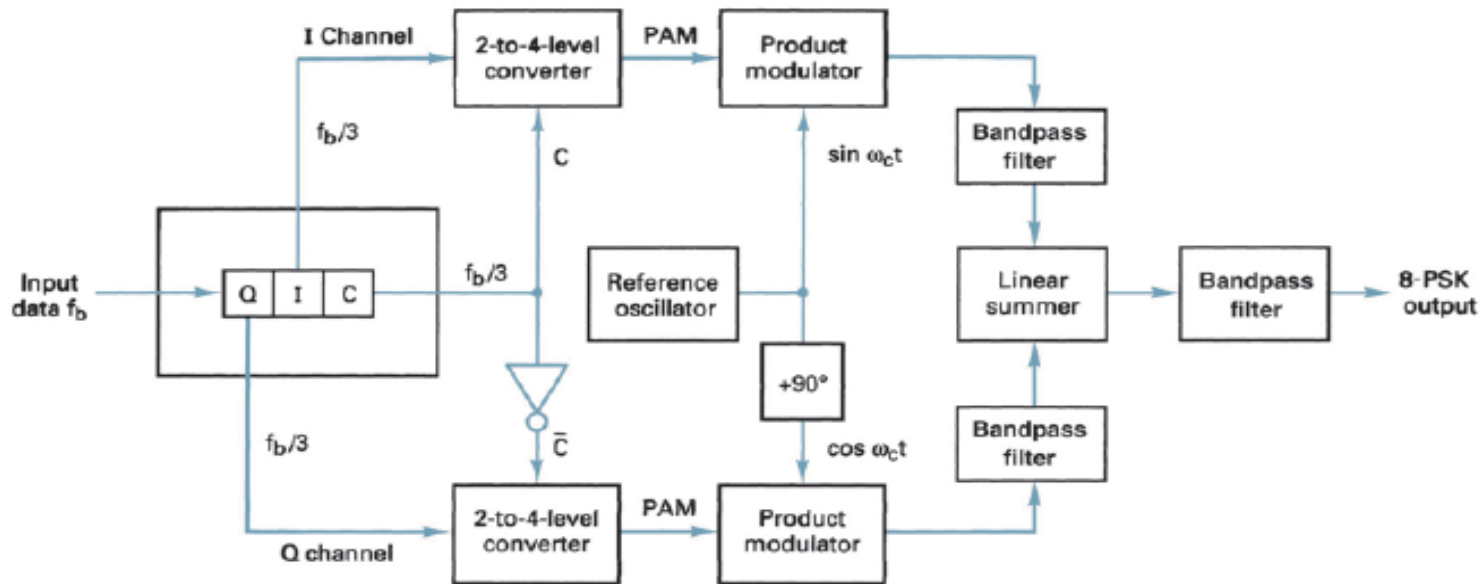


Fig:8-PSK modulator

- The I or Q bit determines the polarity of the output analog signal (logic 1 = +V) and logic 0 = -V), whereas the C or $\bar{C}$ bit determines the magnitude (logic 1 = 1.307 V and logic 0 = 0.541 V).

8-PSK CONT'D

I	C	Output
0	0	-0.541 V
0	1	-1.307 V
1	0	+0.541 V
1	1	+1.307 V

(a)

Q	$\bar{C}$	Output
0	1	-1.307 V
0	0	-0.541 V
1	1	+1.307 V
1	0	+0.541 V

(b)

Fig: I- and Q-channel 2-to-4-level converters: (a) I-channel truth table; (b) Q-channel truth table

- **Example 7:** For a tribit input of $Q=0$, $I=0$, and $C=0$ (000), determine the output phase for the 8-PSK modulator

Solution: The inputs to the I channel 2-to-4-level converter are $I=0$ and $C=0$ and the output is -0.541 V. The inputs to the Q channel 2-to-4-level converter are $Q=0$ and $\bar{C}=1$ and the output is -1.307 V.

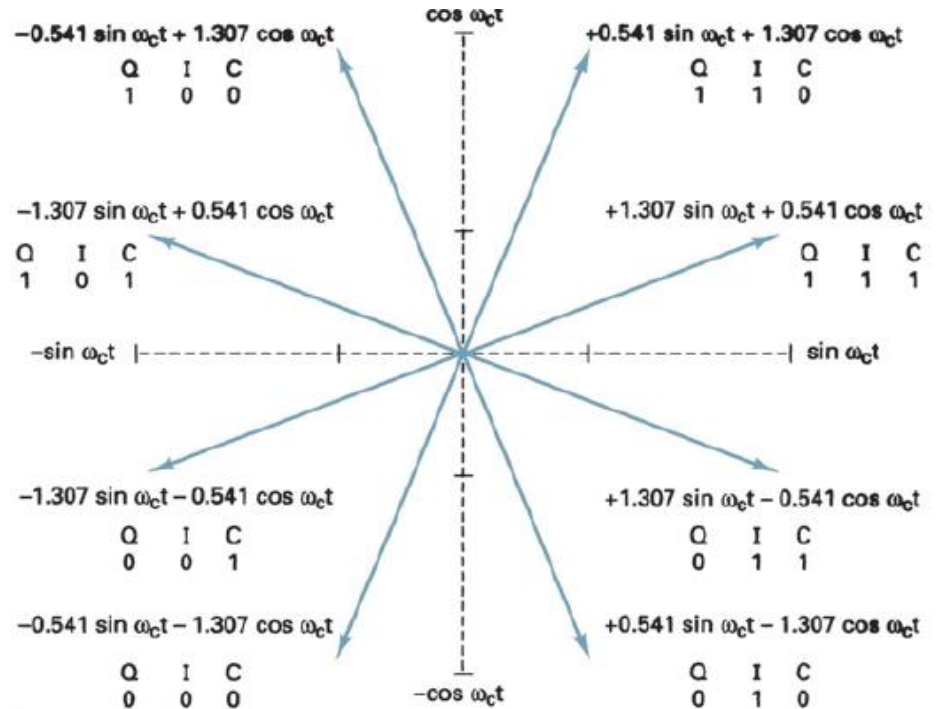
8-PSK CONT'D

- Thus, the two inputs to the I channel product modulators are -0.541 and $\sin \omega_c t$.
The output is $I = (-0.541)(\sin \omega_c t) = -0.541 \sin \omega_c t$
 - The two inputs to the Q channel product modulator are -1.307 V and $\cos \omega_c t$.
The output is $Q = (-1.307)(\cos \omega_c t) = -1.307 \cos \omega_c t$
 - The outputs of the I and Q channel product modulators are combined in the linear summer and produce a modulated output of summer output
- $$\begin{aligned} &= -0.541 \sin \omega_c t - 1.307 \cos \omega_c t \\ &= \sqrt{2} \left[-\frac{0.541}{\sqrt{2}} \sin \omega_c t - \frac{1.307}{\sqrt{2}} \cos \omega_c t \right] \\ &= \sqrt{2} [\sin \omega_c t \cos 112.5^\circ - \sin 112.5^\circ \cos \omega_c t] \\ &= 1.41 \sin(\omega_c t - 112.5^\circ) \end{aligned}$$

8-PSK CONT'D

Binary input			8-PSK output phase
Q	I	C	
0	0	0	-112.5°
0	0	1	-157.5°
0	1	0	-67.5°
0	1	1	-22.5°
1	0	0	$+112.5^\circ$
1	0	1	$+157.5^\circ$
1	1	0	$+67.5^\circ$
1	1	1	$+22.5^\circ$

(a)



(b)

Fig: 8-PSK modulator: (a) truth table; (b) phasor diagram

8-PSK CONT'D

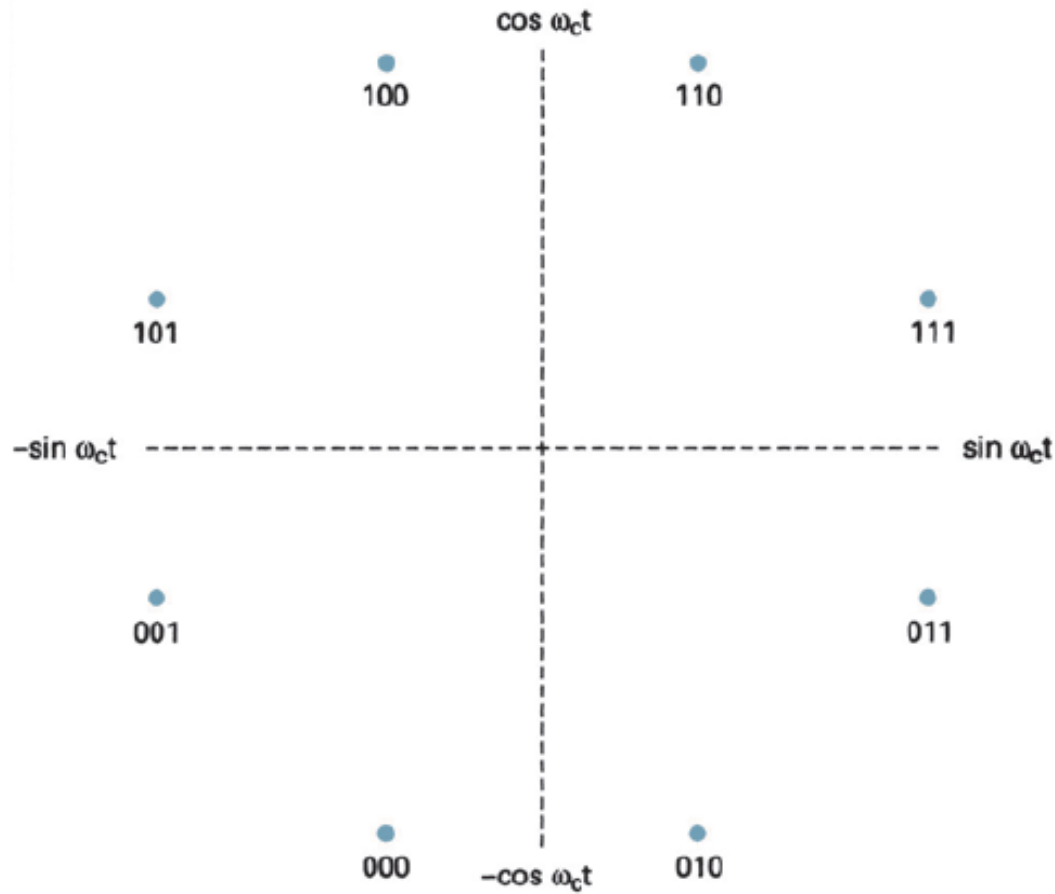
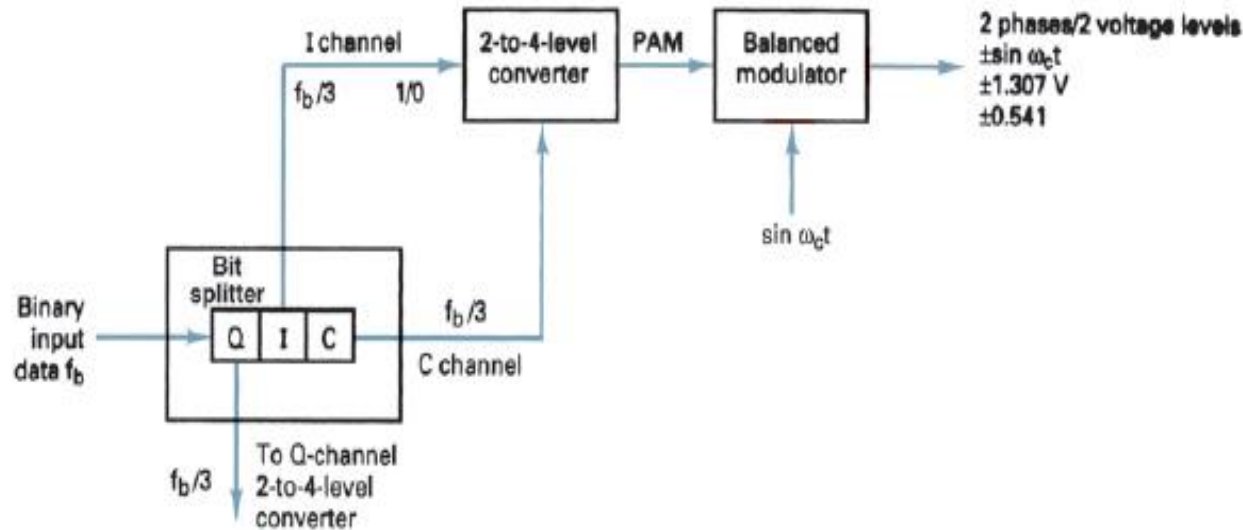


Fig: 8-PSK modulator: constellation diagram

8-PSK CONT'D

Bandwidth considerations of 8-PSK:



Mathematically, the output of the balanced modulators is $\theta = (X \sin \omega_a t)(\sin \omega_c t)$
 $= (X \sin 2\pi f_a t)(\sin 2\pi f_c t)$ [Here, $X = \pm 1.307$ or ± 0.541]

$$= \frac{X}{2} [\cos 2\pi(f_c - f_a)t - \cos 2\pi(f_c + f_a)t]$$

Minimum bandwidth is $(f_c + f_a) - (f_c - f_a) = 2f_a$

$$= 2 \frac{f_b}{3} \quad [\text{N.B: } f_a = \frac{\text{Bit rate}}{2}]$$

$$= \frac{f_b}{3}$$

8-PSK CONT'D

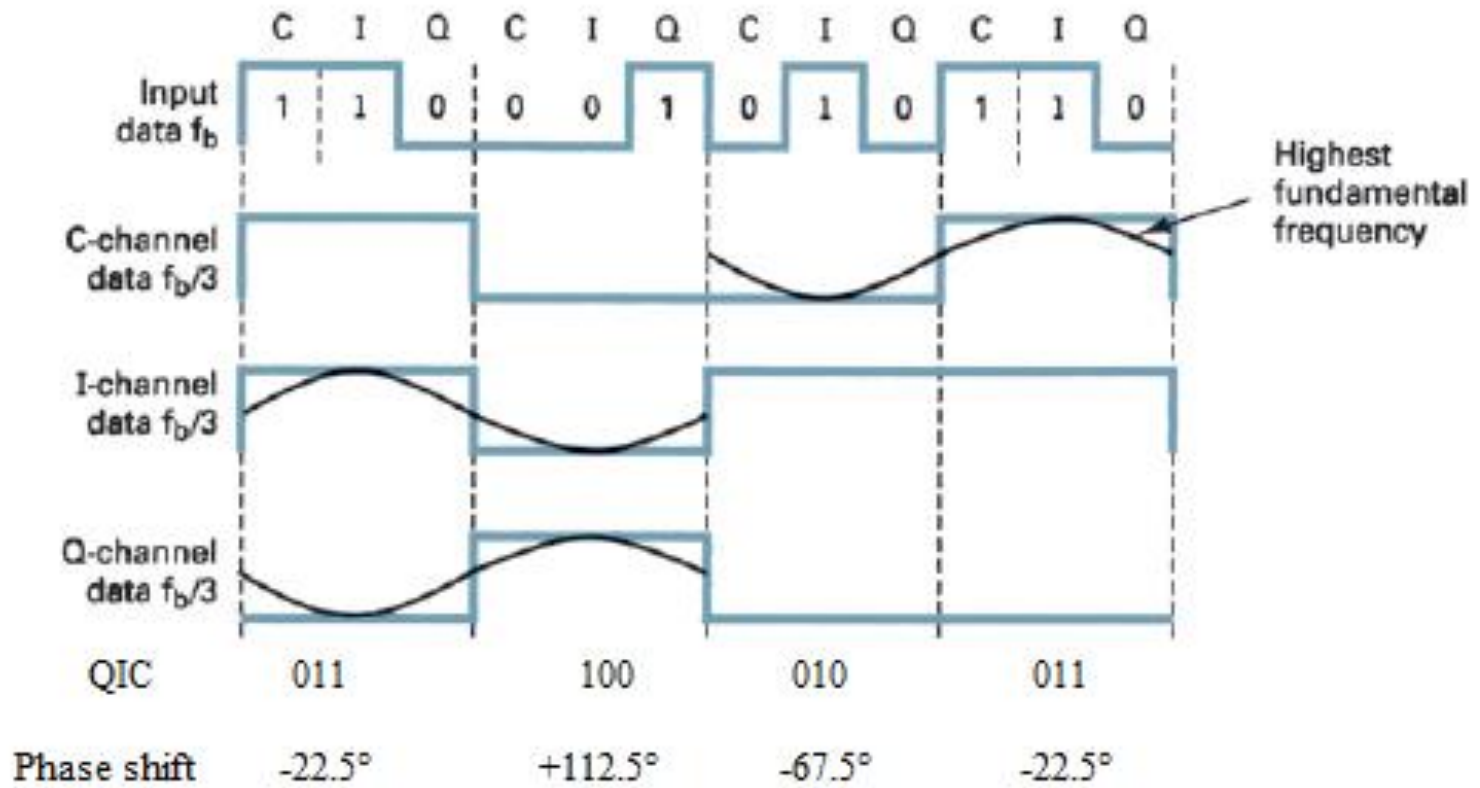


Fig: Timing diagram of 8-PSK

QUADRATURE AMPLITUDE MODULATION(8-QAM)

- With QAM, amplitude and phase-shift keying are combined in such a way that the positions of the signaling elements on the constellation diagrams are optimized to achieve the greatest distance between elements, thus reducing the likelihood of one element being misinterpreted as another element.

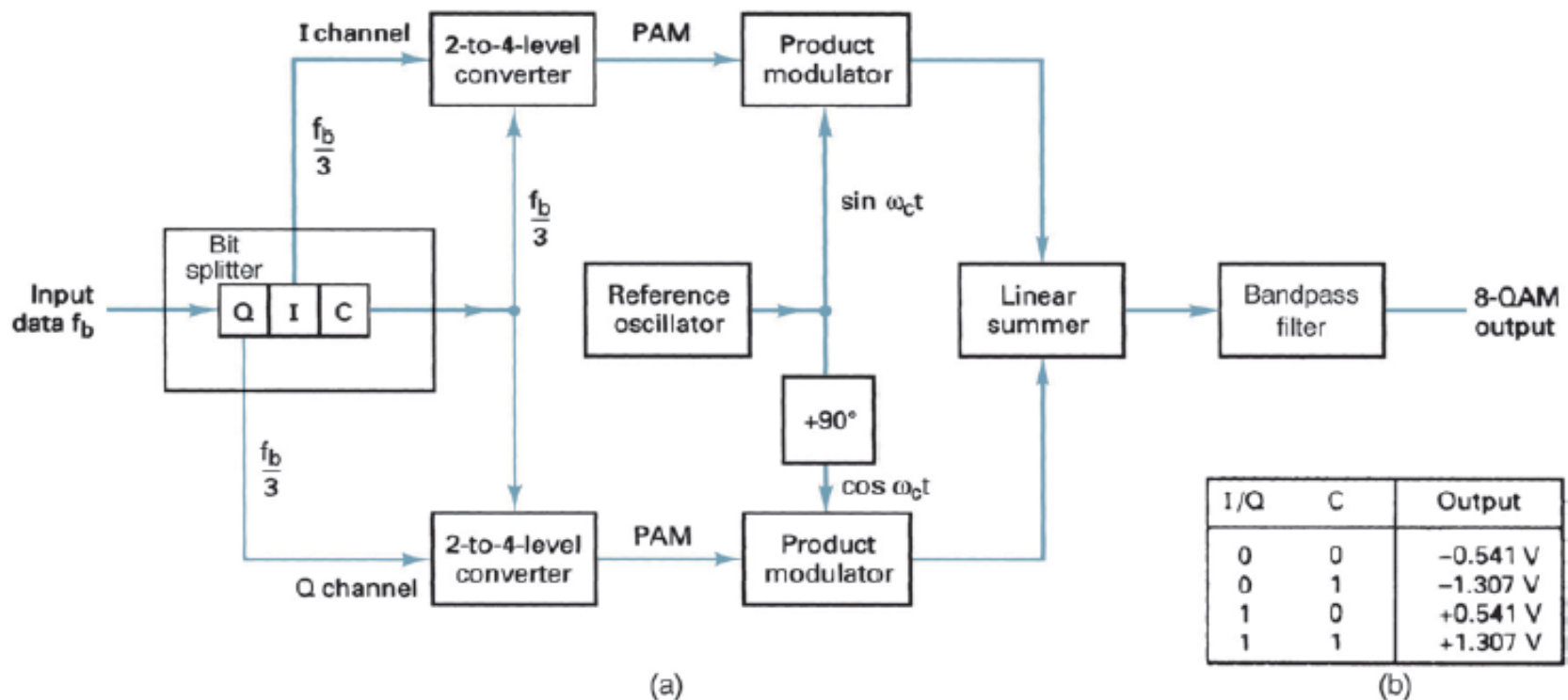


Fig: 8-QAM transmitter: (a) block diagram; (b) truth table 4 level converters

QUADRATURE AMPLITUDE MODULATION(8-QAM) CONT'D

Example 9: For a tribit input of $Q = 0$, $I = 0$, and $C = 0$ (000), determine the output amplitude and phase for the 8-QAM transmitter.

- **Solution:** The inputs to the I channel 2-to-4-level converter are $I = 0$ and $C = 0$ and the output is -0.541 V. The inputs to the Q channel 2-to-4-level converter are $Q = 0$ and $C = 0$ and the output is -0.541 V.
- Thus, the two inputs to the I channel product modulator are -0.541 and $\sin \omega_c t$. The output is $I = (-0.541)(\sin \omega_c t) = -0.541 \sin \omega_c t$
- The two inputs to the Q channel product modulator are -0.541 and $\cos \omega_c t$. The output is $Q = (-0.541)(\cos \omega_c t) = -0.541 \cos \omega_c t$
- The outputs from the I and Q channel product modulators are combined in the linear summer and produce a modulated output of summer output

$$= -0.541 \sin \omega_c t - 0.541 \cos \omega_c t$$

$$= 0.765 \left[-\frac{0.541}{0.765} \sin \omega_c t - \frac{0.541}{0.765} \cos \omega_c t \right] \quad [\text{N.B: } \sqrt{(-0.541)^2 + (-0.541)^2} = 0.765]$$

$$= 0.765 [\sin \omega_c t * \cos 135^\circ - \sin 135^\circ * \cos \omega_c t]$$

$$= 0.765 \sin(\omega_c t - 135^\circ)$$

QUADRATURE AMPLITUDE MODULATION(8-QAM) CONT'D

Another Example: For a tritbit input of $Q = 0$, $I = 1$, and $C = 1$ (011), determine the output amplitude and phase for the 8-QAM transmitter.

- **Solution:** The inputs to the I channel 2-to-4-level converter are $I = 1$ and $C = 1$ and the output is $+1.307$ V. The inputs to the Q channel 2-to-4-level converter are $Q = 0$ and $C = 1$ and the output is -1.307 V.
- Thus, the two inputs to the I channel product modulator are $+1.307$ and $\sin \omega_c t$. The output is $I = (+1.307)(\sin \omega_c t) = +1.307 \sin \omega_c t$
- The two inputs to the Q channel product modulator are -1.307 and $\cos \omega_c t$. The output is $Q = (-1.307)(\cos \omega_c t) = -1.307 \cos \omega_c t$
- The outputs from the I and Q channel product modulators are combined in the linear summer and produce a modulated output of summer output

$$= 1.307 \sin \omega_c t - 1.307 \cos \omega_c t$$

$$= 1.848 \left[\frac{1.307}{1.848} \sin \omega_c t - \frac{1.307}{1.848} \cos \omega_c t \right] \quad [\text{N.B.: } \sqrt{(1.307)^2 + (-1.307)^2} = 1.848]$$

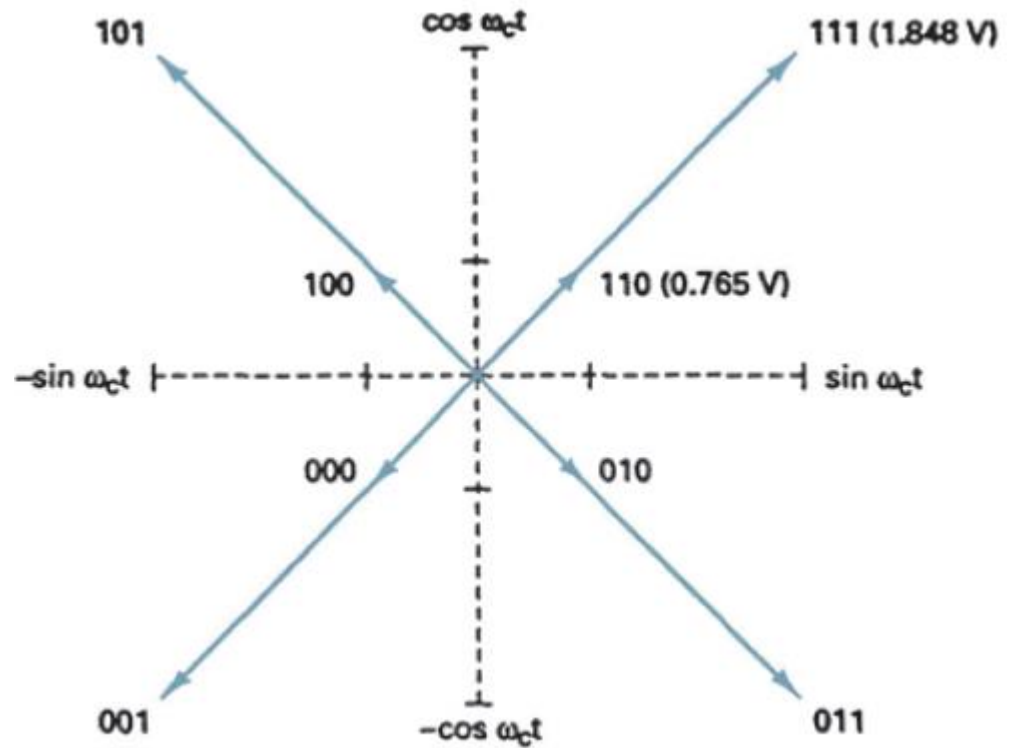
$$= 1.848 [\sin \omega_c t * \cos 45^\circ - \sin 45^\circ * \cos \omega_c t]$$

$$= 1.848 \sin(\omega_c t - 45^\circ)$$

QUADRATURE AMPLITUDE MODULATION(8-QAM) CONT'D

Binary input			8-QAM output	
Q	I	C	Amplitude	Phase
0	0	0	0.765 V	-135°
0	0	1	1.848 V	-135°
0	1	0	0.765 V	-45°
0	1	1	1.848 V	-45°
1	0	0	0.765 V	$+135^\circ$
1	0	1	1.848 V	$+135^\circ$
1	1	0	0.765 V	$+45^\circ$
1	1	1	1.848 V	$+45^\circ$

(a)



(b)

Fig: 8-QAM modulator: (a) truth table; (b) phasor diagram

QUADRATURE AMPLITUDE MODULATION(8-QAM) CONT'D

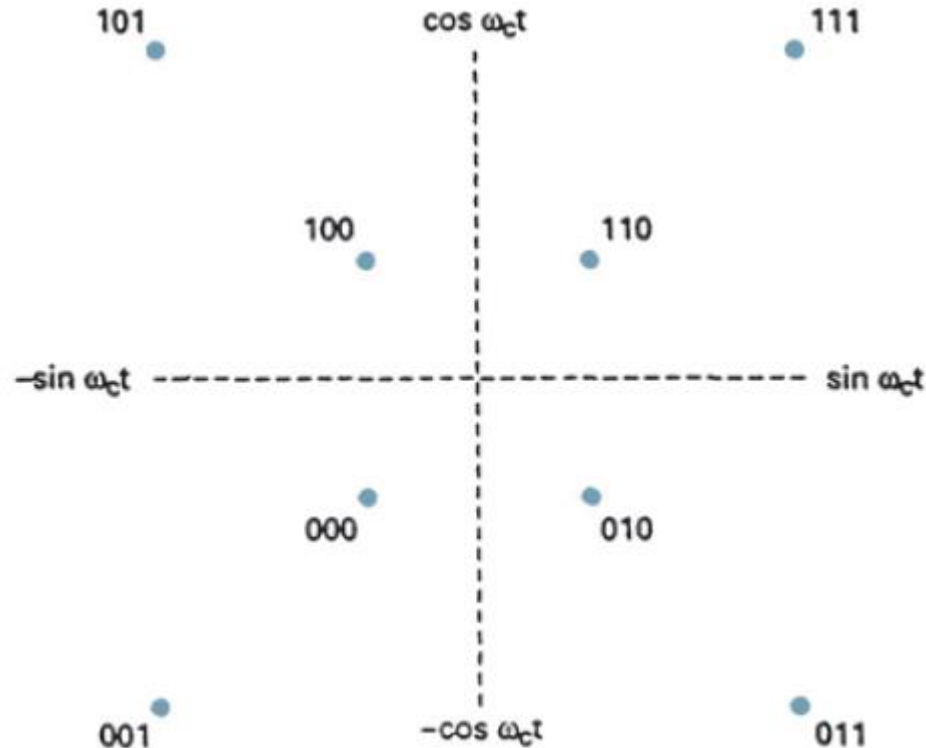


Fig: 8-QAM modulator: constellation diagram

- **Bandwidth considerations of 8-QAM:** Minimum bandwidth is $\frac{f_b}{3}$

[Same as 8-PSK]

BIT RATE & BAUD

- **Bit rate:** is simply the number of bits transmitted during one second and is expressed in bits per second (bps).
- **Baud:** refers to the rate of change of a signal on the transmission medium after encoding and modulation have occurred. Mathematically, baud is the reciprocal of the time of one output signaling element (also called symbol), and a signaling element may represent several information bits. Baud is expressed as
$$\text{baud} = \frac{1}{t_s}$$

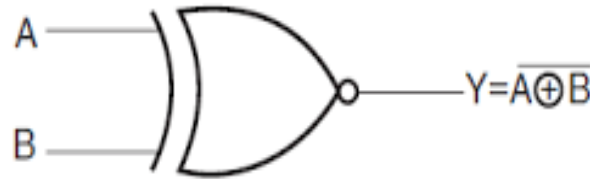
Where, baud = symbol rate (baud per second)

t_s = time of one signaling element (seconds)

- Baud also equals the bit rate divided by the number of bits encoded into one signaling element. Thus,
$$\text{baud} = \frac{f_b}{N}$$
- In binary systems (such as binary FSK and binary PSK), baud and bits per second are equal. However, in higher-level systems (such as QPSK and 8-PSK), bps is always greater than baud.
- **Example:** 1, 2, 3, 4, 6, 8.

DIFFERENTIAL PHASE SHIFT KEYING (DPSK)

- DPSK is an alternative form of digital modulation where the binary input information is contained in the difference between two successive signaling elements rather than the absolute phase. With DPSK, it is not necessary to recover a phase-coherent carrier.



$$Y = \overline{(A \oplus B)} = (A.B + \overline{A}.\overline{B})$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

DIFFERENTIAL PHASE SHIFT KEYING (DPSK) CONT'D

❖ DBPSK transmitter:

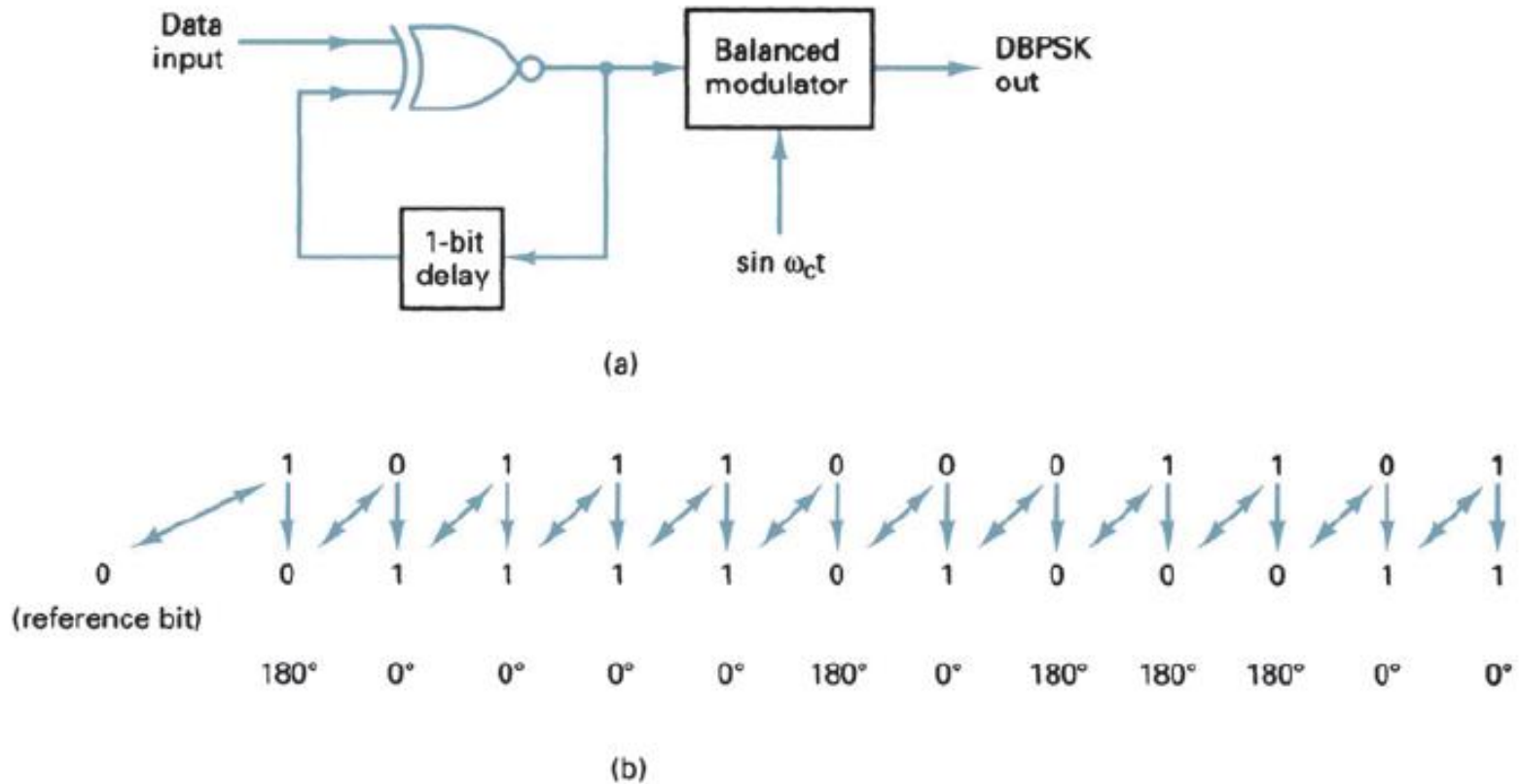


Fig: DBPSK modulator: (a) block diagram; (b) timing diagram

DIFFERENTIAL PHASE SHIFT KEYING (DPSK) CONT'D

- An incoming information bit is XNORed with the preceding bit prior to entering the BPSK modulator (balanced modulator).
- For the first data bit, there is no preceding bit with which to compare it. Therefore, an initial reference bit is assumed.
- If the initial reference bit is assumed a logic 1, the output from the XNOR circuit is simply the complement of that shown.
- The first data bit is XNORed with the reference bit. If they are the same, the XNOR output is a logic 1; if they are different, the XNOR output is a logic 0.
- The balanced modulator operates the same as a conventional BPSK modulator; a logic 1 produces in $+\sin \omega_c t$ at the output, and a logic 0 produces $-\sin \omega_c t$ at the output.

DIFFERENTIAL PHASE SHIFT KEYING (DPSK) CONT'D

❖ DBPSK receiver:

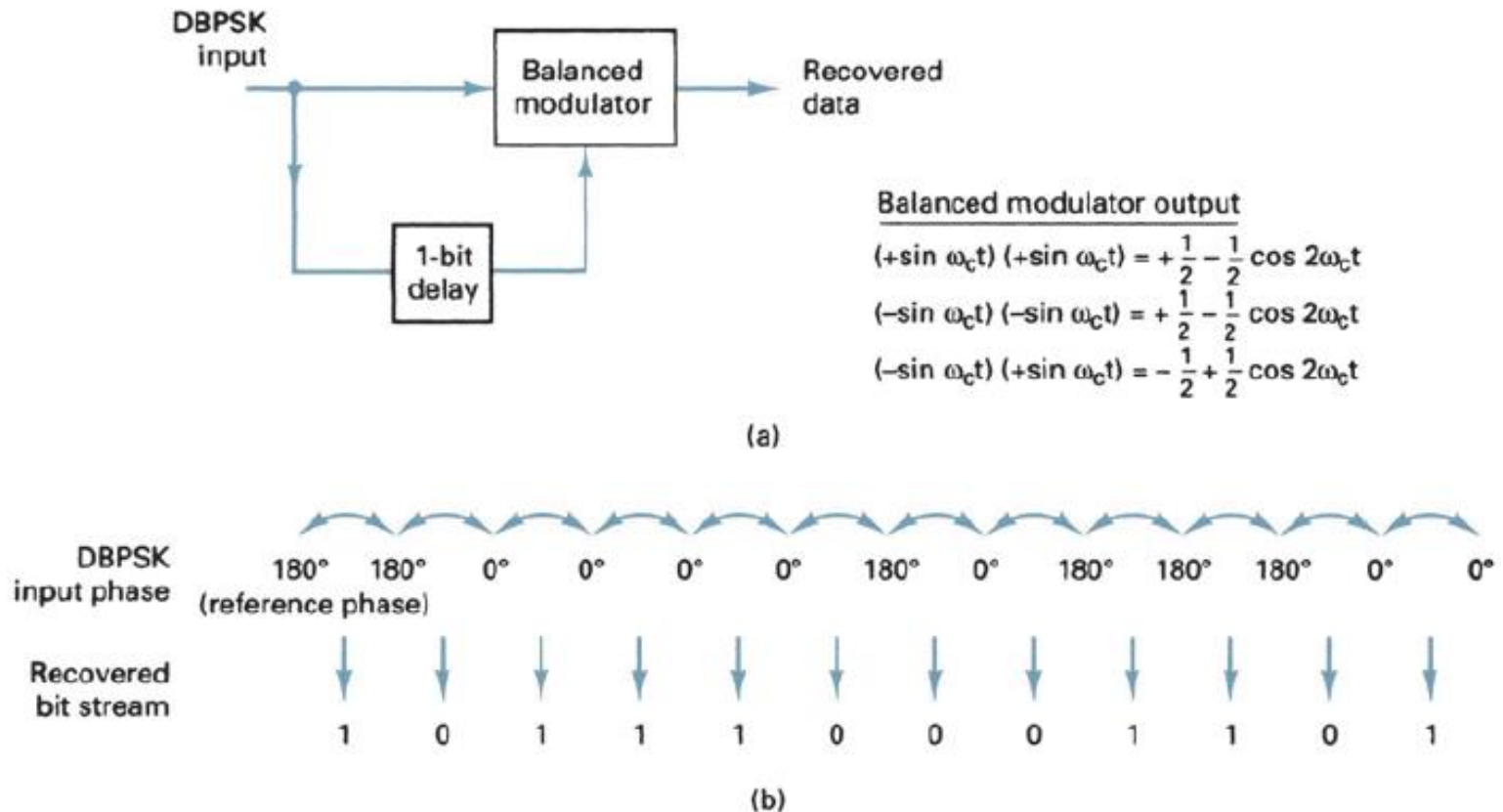


Fig: DBPSK demodulator: (a) block diagram; (b) timing sequence

DIFFERENTIAL PHASE SHIFT KEYING (DPSK) CONT'D

- The received signal is delayed by one bit time, then compared with the next signaling element in the balanced modulator.
- If they are the same, a logic 1 (+ voltage) is generated. If they are different, a logic 0 (- voltage) is generated.
- If the reference phase is incorrectly assumed, only the first demodulated bit is in error.
- The primary advantage of DBPSK is the simplicity with which it can be implemented. With DBPSK, no carrier recovery circuit is needed.
- A disadvantage of DBPSK is that it requires between 1 dB and 3 dB more signal-to-noise ratio to achieve the same bit error rate as that of absolute PSK.